

EMPLOYEE DATABASE

```
show databases;
```

```
use newdatabase;
```

```
create database emp;
```

```
use emp;
```

```
CREATE TABLE dept (  
    deptno decimal(2,0) primary key,  
    dname varchar(14) default NULL,  
    loc varchar(13) default NULL  
);
```

```
CREATE TABLE emp (  
    empno decimal(4,0) primary key,  
    ename varchar(10) default NULL,  
    mgr_no decimal(4,0) default NULL,  
    hiredate date default NULL,  
    sal decimal(7,2) default NULL,  
    deptno decimal(2,0) references dept(deptno) on delete cascade on update cascade  
);
```

```
create table incentives (  
    empno decimal(4,0) references emp(empno) on delete cascade on update cascade,  
    incentive_date date,
```

```
incentive_amount decimal(10,2),  
primary key(empno,incentive_date)  
);
```

```
Create table project (  
pno int primary key,  
pname varchar(30) not null,  
ploc varchar(30)  
);
```

```
Create table assigned_to (  
empno decimal(4,0) references emp(empno) on delete cascade on update cascade,  
pno int references project(pno) on delete cascade on update cascade,  
job_role varchar(30),  
primary key(empno,pno)  
);
```

```
show tables;
```

```
INSERT INTO dept VALUES (10,'ACCOUNTING','MUMBAI');  
INSERT INTO dept VALUES (20,'RESEARCH','BENGALURU');  
INSERT INTO dept VALUES (30,'SALES','DELHI');  
INSERT INTO dept VALUES (40,'OPERATIONS','CHENNAI');
```

```
INSERT INTO emp VALUES (7369,'Adarsh',7902,'2012-12-17','80000.00','20');  
INSERT INTO emp VALUES (7499,'Shruthi',7698,'2013-02-20','16000.00','30');  
INSERT INTO emp VALUES (7521,'Anvitha',7698,'2015-02-22','12500.00','30');  
INSERT INTO emp VALUES (7566,'Tanvir',7839,'2008-04-02','29750.00','20');  
INSERT INTO emp VALUES (7654,'Ramesh',7698,'2014-09-28','12500.00','30');  
INSERT INTO emp VALUES (7698,'Kumar',7839,'2015-05-01','28500.00','30');  
INSERT INTO emp VALUES (7782,'CLARK',7839,'2017-06-09','24500.00','10');
```

```
INSERT INTO emp VALUES (7788,'SCOTT',7566,'2010-12-09','30000.00','20');
INSERT INTO emp VALUES ('7839','KING',NULL,'2009-11-17','500000.00','10');
INSERT INTO emp VALUES ('7844','TURNER',7698,'2010-09-08','15000.00','30');
INSERT INTO emp VALUES ('7876','ADAMS',7788,'2013-01-12','11000.00','20');
INSERT INTO emp VALUES ('7900','JAMES',7698,'2017-12-03','9500.00','30');
INSERT INTO emp VALUES ('7902','FORD','7566','2010-12-03','30000.00','20');
```

```
INSERT INTO incentives VALUES(7499,'2019-02-01',5000.00);
INSERT INTO incentives VALUES(7521,'2019-03-01',2500.00);
INSERT INTO incentives VALUES(7566,'2022-02-01',5070.00);
INSERT INTO incentives VALUES(7654,'2020-02-01',2000.00);
INSERT INTO incentives VALUES(7654,'2022-04-01',879.00);
INSERT INTO incentives VALUES(7521,'2019-02-01',8000.00);
INSERT INTO incentives VALUES(7698,'2019-03-01',500.00);
INSERT INTO incentives VALUES(7698,'2020-03-01',9000.00);
INSERT INTO incentives VALUES(7698,'2022-04-01',4500.00);
```

```
INSERT INTO project VALUES(101,'AI Project','BENGALURU');
INSERT INTO project VALUES(102,'IOT','HYDERABAD');
INSERT INTO project VALUES(103,'BLOCKCHAIN','BENGALURU');
INSERT INTO project VALUES(104,'DATA SCIENCE','MYSURU');
INSERT INTO project VALUES(105,'AUTONOMUS SYSTEMS','PUNE');
```

```
INSERT INTO assigned_to VALUES(7499,101,'Software Engineer');
INSERT INTO assigned_to VALUES(7521,101,'Software Architect');
INSERT INTO assigned_to VALUES(7566,101,'Project Manager');
INSERT INTO assigned_to VALUES(7654,102,'Sales');
INSERT INTO assigned_to VALUES(7521,102,'Software Engineer');
INSERT INTO assigned_to VALUES(7499,102,'Software Engineer');
INSERT INTO assigned_to VALUES(7654,103,'Cyber Security');
```

```
INSERT INTO assigned_to VALUES(7698,104,'Software Engineer');
```

```
INSERT INTO assigned_to VALUES(7900,105,'Software Engineer');
```

```
INSERT INTO assigned_to VALUES(7839,104,'General Manager');
```

```
Select * from dept;
```

```
select * from emp;
```

```
select * from incentives;
```

```
select * from project;
```

```
select * from assigned_to;
```

#1 question

```
SELECT emp.empno
```

```
FROM emp
```

```
LEFT JOIN incentives ON emp.empno = incentives.empno
```

```
WHERE incentives.empno IS NULL;
```

#2 question

```
SELECT m.ename, count(*)
```

```
FROM emp e,emp m
```

```
WHERE e.mgr_no = m.empno
```

```
GROUP BY m.ename
```

```
HAVING count(*) =(SELECT MAX(mycount)
```

```
from (SELECT COUNT(*) mycount
```

```
FROM emp
```

```
GROUP BY mgr_no) a);
```

#3 question

```
SELECT m.empno AS manager_id, AVG(e.sal) AS avg_salary
FROM emp m
JOIN emp e ON e.mgr_no = m.empno
GROUP BY m.empno;
```

#4 question

```
SELECT DISTINCT m.mgr_no
FROM emp e, emp m
WHERE e.mgr_no = m.mgr_no
AND e.deptno = m.deptno
AND e.empno IN (
    SELECT DISTINCT e2.mgr_no
    FROM emp e2
    WHERE e2.mgr_no IS NOT NULL
);
```

IMAGES

The image displays two screenshots of the MySQL Workbench interface, showing SQL queries and their results.

Top Screenshot:

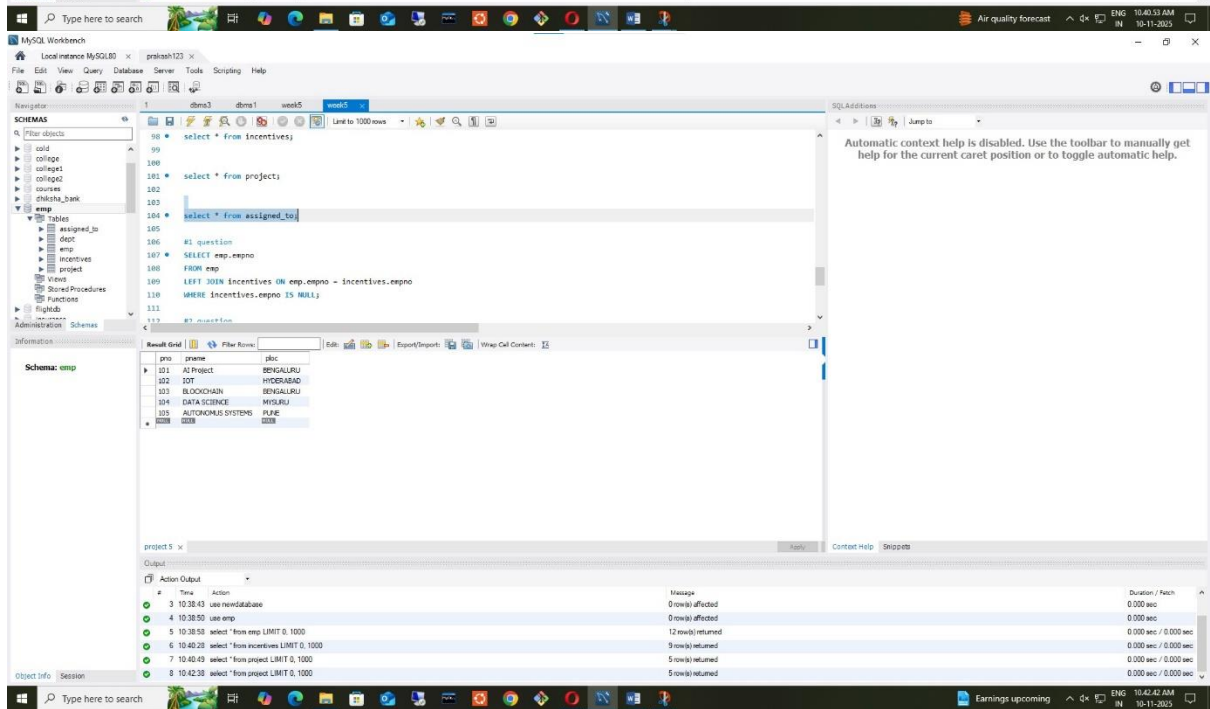
- Navigator:** The 'emp' schema is selected, showing tables like 'assigned_to', 'dept', 'emp', 'incentives', 'project', 'views', 'shared_procedures', 'functions', and 'triggers'.
- SQL Editor:** The query is:

```
90 * INSERT INTO assigned_to VALUES(7499,184,'Software Engineer');
91 * INSERT INTO assigned_to VALUES(7654,184,'Cyber Security');
92 * INSERT INTO assigned_to VALUES(7654,184,'Software Engineer');
93 * INSERT INTO assigned_to VALUES(7654,184,'General Manager');
94 * Select * from dept;
```
- Result Grid:** The query results are displayed in a table with columns: empno, ename, mgr_no, hiredate, sal, deptno. The results show a list of employees and their details.
- Output:** The output shows the execution of the queries, including the number of rows affected and the duration of each query.

Bottom Screenshot:

- Navigator:** The 'emp' schema is selected, showing tables like 'assigned_to', 'dept', 'emp', 'incentives', 'project', 'views', 'shared_procedures', 'functions', and 'triggers'.
- SQL Editor:** The query is:

```
90 * INSERT INTO assigned_to VALUES(7499,184,'Software Engineer');
91 * INSERT INTO assigned_to VALUES(7654,184,'Cyber Security');
92 * INSERT INTO assigned_to VALUES(7654,184,'Software Engineer');
93 * INSERT INTO assigned_to VALUES(7654,184,'General Manager');
94 * Select * from dept;
```
- Result Grid:** The query results are displayed in a table with columns: empno, ename, mgr_no, hiredate, sal, deptno. The results show a list of employees and their details.
- Output:** The output shows the execution of the queries, including the number of rows affected and the duration of each query.



MySQL Workbench

Local instance MySQL80 x prakash123 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

- gold
- colleg1
- colleg2
- colleg4
- colleg5
- courses
- chiksha_bank
- emp
 - tables
 - assigned_to
 - dept
 - emp
 - incentives
 - project
 - views
 - Stored Procedures
 - Functions
 - Right-click

Information

Schema: emp

Result Grid

Filter Rows: Limit to 1000 rows

Exports | Wrap Cell Contents

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

1187 SELECT emp.empno
1188 FROM emp
1189 LEFT JOIN incentives ON emp.empno = incentives.empno
1190 WHERE incentives.empno IS NULL;
1191
1192 #2 question
1193 SELECT m.ename, count(*)
1194 FROM emp e, emp m
1195 WHERE e.mgr_no = m.empno
1196 GROUP BY m.ename
1197 HAVING count(*) > (SELECT MAX(mycount)
1198 FROM (SELECT count(*) mycount
1199 FROM emp
1200 GROUP BY mgr_no) a);
1201

```

Result 7

empno	count(*)
1000	5

Output

Action Output

#	Time	Action	Message	Duration / Feat
5	10:38:53	select * from emp LIMIT 0, 1000	12 rows returned	0.000 sec / 0.000 sec
6	10:40:28	select * from incentives LIMIT 0, 1000	9 rows returned	0.000 sec / 0.000 sec
7	10:40:43	select * from project LIMIT 0, 1000	5 rows returned	0.000 sec / 0.000 sec
8	10:42:38	select * from emp LEFT JOIN incentives ON emp.empno = incentives.empno WHERE incentives.empno IS NULL LIMIT 0, 1000	7 rows returned	0.000 sec / 0.000 sec
9	10:43:21	SELECT emp.empno FROM emp LEFT JOIN incentives ON emp.empno = incentives.empno WHERE incentives.empno IS NULL LIMIT 0, 1000	7 rows returned	0.000 sec / 0.000 sec
10	10:43:50	SELECT m.ename, count(*) FROM emp e, emp m WHERE e.mgr_no = m.empno GROUP BY m.ename HAVING count(*) > (SELECT MAX(mycount) FROM (SELECT count(*) mycount FROM emp GROUP BY mgr_no) a);	1 rows returned	0.000 sec / 0.000 sec

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Schema: emp

Result Grid

Filter Rows: Limit to 1000 rows

Exports | Wrap Cell Contents

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```

1213 SELECT m.ename, count(*)
1214 FROM emp e, emp m
1215 WHERE e.mgr_no = m.empno
1216 GROUP BY m.ename
1217 HAVING count(*) > (SELECT MAX(mycount)
1218 FROM (SELECT count(*) mycount
1219 FROM emp
1220 GROUP BY mgr_no) a);
1221
1222 #3 question
1223 SELECT m.empno AS manager_id, MIN(e.sal) AS avg_salary
1224 FROM emp m
1225 JOIN emp e ON e.mgr_no = m.empno
1226 GROUP BY m.empno

```

Result 8

manager_id	avg_salary
7902	8000.000000
7969	12000.000000
7966	30000.000000
7788	11000.000000

Output

Action Output

#	Time	Action	Message	Duration / Feat
6	10:40:28	select * from incentives LIMIT 0, 1000	9 rows returned	0.000 sec / 0.000 sec
7	10:40:43	select * from project LIMIT 0, 1000	5 rows returned	0.000 sec / 0.000 sec
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9	10:43:21	SELECT emp.empno FROM emp LEFT JOIN incentives ON emp.empno = incentives.empno WHERE incentives.empno IS NULL LIMIT 0, 1000	7 rows returned	0.000 sec / 0.000 sec
10	10:43:50	SELECT m.ename, count(*) FROM emp e, emp m WHERE e.mgr_no = m.empno GROUP BY m.ename HAVING count(*) > (SELECT MAX(mycount) FROM (SELECT count(*) mycount FROM emp GROUP BY mgr_no) a);	1 rows returned	0.000 sec / 0.000 sec
11	10:44:11	SELECT m.empno AS manager_id, AVG(e.sal) AS avg_salary FROM emp m JOIN emp e ON e.mgr_no = m.empno GROUP BY m.empno LIMIT 0, 1000	4 rows returned	0.000 sec / 0.000 sec